Notes on the Structural Macro Policy Simulation Model

GIZ - MINECOFIN Rwanda

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Main properties of the model

- Quarterly model of a small open economy for fiscal projections and simulations
- Captures the dominance of fiscal policy in Rwanda
  - transfers to hand-to-mouth households
  - weak transmission due to liquidity constrained households
- Monetary policy is a combination of inflation and exchange rate targeting
- Theoretically, the model combines features of semi-structural gap models and reduced form DSGE models
- There’s partial stock-flow consistency, making the model non-linear

Parallel decompositions

- Fiscal balance: revenues/expenditures vs. cyclical/structural

$$FB_t = T_t - (G_t + IG_t) + NFG_t = \overline{FB}_t + \tilde{FB}_t \tag{1}$$

- GDP: demand side vs. trend/gap:

$$Y_t = C_t + I_t + G_t + X_t - M_t = \overline{Y}_t + \tilde{Y}_t \tag{2}$$

- CPI: core/food/energy vs. domestic/imported

$$\pi_t^{\text{cpi}} = \pi_t^{\text{core}} + \pi_t^{\text{food}} + \pi_t^{\text{ener}} = \pi_t^{\text{dom}} + \pi_t^{\text{imp}} \tag{3}$$

- With such parallel decompositions, one variable must always be given as a residual

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Policy channels

- A consequence is that both fiscal and monetary policy affect the economy through parallel channels
- Fiscal policy:
  - through the effect of disposable income on consumption
  - through the fiscal impulse in the output gap equations
- Monetary policy:
  - through the *level* of real interest rates in the consumption and investment equations
  - through the real interest rate *gap* in the output gap equations

Balanced growth path

- Gap models automatically (almost) have a BGP: trends are exogenous
- In reduced form structural models we usually need parameter restrictions
- Simple rule-of-thumb equations have to be added carefully
- Example
  - We'd like disposable income and consumption to grow with GDP in the steady state
  - This implies $\tau_1 = 1$ in

$$\log T_t = \tau_1 \log Y_t + \tau_0 \tag{4}$$

  - And then we must have $\gamma_1 + \gamma_3 + \gamma_5 = 1$ in

$$\log C_t = \gamma_1 \log C_{t-1} + \gamma_2 RR_t + \gamma_3 \log YD_t + \gamma_4 RR_{t-1} + \gamma_5 \log YD_{t-1} \tag{5}$$

with $YD_t = Y_t - T_t$ and RR constant in the steady state

RIR and RER definitions

- The decision whether to defined the real interest rate and the real exchange rate in terms of core or headline CPI is not always innocuous
- Especially not when the steady states of food and energy inflation are different from core
- Introducing separate definitions of the real exchange rate might jeopardise its interpretation as a indicator of convergence
- Trends of relative prices can capture all these ”uncomfortable” (from the perspective of model operation) long term discrepancies
- And relative price gaps can be incorporated whenever the economically meaningful short term dynamics calls for it

The production function

- Potential output is given by a production function
  - And also by an AR(1) process???
  - Or maybe an HP filter???
- The only endogenous factor is capital
  - Or is it given by a linear trend???
- The role of the production function through optimal investment decisions and the equilibrium on the goods market – which is a key mechanism of DSGE models –, is completely missing from the reduced form model
- The value added of this formulation is therefore not clear

Shocks

- Shocks are not specified
- It is very important to have the correct shock specification
  - Too many shocks would make their economic content unclear, filter results would be hard to interpret
  - Too few shocks would hinder forecasting (i.e. starting from the actual initial conditions)

Proposed modifications: auxiliary model

- Move all ”reporting equations” to a separate model
- Money, deficit financing, external balance
- These variables have no feedback to the rest of the model
- Estimation, filtering and simulations would be faster

Proposed modifications: production function

- Model potential output as an exogenous trend, i.e. remove the capital accumulation equation
- Not really relevant on the 12-16 quarters horizon
- Capital used in production depends on a lot of other variables
  - Variations in the depreciation rate
  - Capacity utilization
- This trend of course can be tuned based on expert knowledge about its determinants (labor, human capital, productivity)

Proposed modifications: GDP identity

- Model GDP as the sum of potential and the output gap
- Move the demand decomposition to the auxiliary model
- Individual demand components do not have a feedback to the core model
  - Inflation depends on the output gap
- Parameters could be estimated by simple OLS
- Adding a discrepancy term, such as inventories, would also be necessary
  - In general equilibrium models this is an equilibrium condition
  - Relative prices adjust to equate demand and supply
  - We don't have such a mechanism or variables, that's why we need a discrepancy term

Proposed modifications: food and energy prices

- Model their relative prices to core
- As the sum of a gap and trend
- The trend is completely exogenous (with possible non-zero steady state change)
- The gap depends on corresponding relative world price gaps (e.g. oil to US CPI)
- This way we can have a core Phillips curve without relative prices

Proposed modifications: fiscal policy

- Formulate all equations in terms of ratios (i.e. in percent of GDP)
- In the dynamic relationships, use CPI inflation
- The difference between the CPI and the GDP deflator would probably cause little discrepancies, given the relatively small share of exports
- Using the CPI as the main deflator makes the GDP deflator a reporting variable, which can be moved to the auxiliary model

Proposed modifications: RIR and RER

- Define the real interest rate and real exchange rate in terms of core prices
- This is the best way to avoid complications with relative prices
- Use relative price gaps directly whenever necessary (e.g. oil price gap in the IS curve)
- Calculate and use an implicit core inflation target to assure a correct steady state
- For reporting, we can calculate CPI based series in the auxiliary model

Proposed modifications: summary

- Start from a standard quarterly gap model, with exogenous trends, FX targeting and hybrid UIP
- Defined the real interest rate and the real exchange rate in terms of core prices
- Describe food and energy prices through their relative price gaps
- Model fiscal policy using the strucutral/cyclical/discretionary decomposition of the deficit ratio
- Add debt and interest payments, and define the fiscal impulse in terms of the primary deficit
- Calculate additional variables in an auxiliary model
  - demand components, deficit financing, money, external balance
  - add discrepancy terms when necessary (e.g. inventories in the GDP identity)

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